
Optimizer Mismatch Drives Catastrophic Forgetting

Celine Tan

University of California, Berkeley
celinetan@berkeley.edu

E Harrison

University of California, Berkeley
ehharrison@berkeley.edu

Katie Wang

University of California, Berkeley
katiewang@berkeley.edu

Swetha Rajkumar

University of California, Berkeley
swetha.rajkumar@berkeley.edu

Ryan Cheng

University of California, Berkeley
ryancheng@berkeley.edu

Abstract

A major challenge in the continual learning setting is catastrophic forgetting, which occurs when training a model on a new task causes a rapid deterioration in its performance on previously learned tasks. We analyze how the combination of different optimizers during pre-training and fine-tuning affects catastrophic forgetting. Instantiating from two variants of Moonlight-16B [10] pre-trained with AdamW and Muon, we investigate how model performance on pre-training tasks evolves when fine-tuning with a different optimizer compared to pre-training, and when fine-tuning with the same optimizer as used in pre-training. We investigate two regimes: (i) *in-distribution* supervised fine-tuning on English web-text continuation [15], where the SFT distribution lies inside the pre-training data mix, and (ii) *out-of-distribution* fine-tuning on autonomous-driving language-action prediction [16][4]. Within the autonomous driving domain, optimizer mismatch consistently worsens catastrophic forgetting under full fine-tuning; We additionally investigate LoRA as a mitigation strategy for forgetting, and measure whether this effect persists despite LoRA. On English web-text continuation, optimizer mismatch still worsens evaluation metrics even under in-distribution fine-tuning, and weight-change analysis reveals that different optimizers follow qualitatively distinct loss geometries even in this setting.

1 Introduction

Continual learning (CL) studies how models adapt to a sequence of tasks without retraining from scratch. A central failure mode is *catastrophic forgetting* [13]: performance on earlier tasks degrades when the model is optimized for a new objective. Fine-tuning a pretrained large language model (LLM) is a special case: pretraining shapes a broad representation, and a short supervised fine-tuning (SFT) phase adapts it to a downstream task. Many methods aim to limit forgetting during this second phase, including constrained optimization [17, 11, 3], regularization [18, 12, 8], and parameter-efficient adapters.

Prior work has compared optimizers such as Adam and SGD in forgetting settings [1, 5, 7]. Recent LLM training practice introduces *Muon* [10], which orthogonalizes updates on 2D weight matrices and can differ substantially from AdamW in implicit regularization. Moonshot reports that *optimizer*

mismatch, e.g., using AdamW in pretraining and Muon in fine-tuning, or vice versa, can hurt downstream quality even when architecture and data are held fixed.

We ask whether this mismatch also implies catastrophic forgetting of general pre-training capabilities when models are adapted to a new task. The answer is not obvious because forgetting depends on both optimization dynamics and how far the fine-tuning distribution lies from pretraining. When fine-tuning data are encompassed by the pretraining distribution (e.g., English web-text continuation in the style of C4) SFT mostly adjusts behavior within a region the model already supports. When fine-tuning is deliberately out of distribution (e.g., language-action prediction for autonomous driving) the optimizer must negotiate a sharper trade-off between new-task performance and retention of general knowledge. We posit that, due to the distinct forms of implicit regularization found in popular optimization methods (e.g. AdamW and Muon), optimizer mismatch between pre-training and fine-tuning exacerbates catastrophic forgetting.

To investigate this phenomenon, we fine-tune Moonlight-16B-A3B [10], using AdamW and Muon-pretrained instantiations with matched architecture and pre-training datasets, and fine-tune each model with AdamW and Muon on two domains of varying adherence to the pre-training regime. We additionally perform mechanistic analysis of the evolution of the weights and loss landscapes of each model over the course of fine-tuning.

2 Related Work

2.1 Catastrophic Forgetting in Large Language Models

Catastrophic forgetting in neural networks has been studied for decades [13, 5], and recent work has come to investigate how new optimizers affect the pretraining of large language models. While most models today still use AdamW as their training optimizer of choice, Muon [10] has been shown to scale to large LLM pretraining, and has led to further research into how different optimizers affect performance. The work from Moonshot AI resulting in *Moonlight* [10] also reported how Muon or Adam pretraining interacted with the choice of fine-tuning optimizer. We extend upon their work by measuring and investigating forgetting on fixed benchmarks under controlled optimizer pairings and two fine-tuning distributions (encompassed vs. OOD).

2.2 Optimizer Implicit Regularization

A key insight developed by [2] is that gradient-based optimizers can be modeled as trying to solve the same constrained optimization question, just over different norms. This works as a form of implicit regularization by the optimizer and makes updates to optimal regions in parameter space dependent on this norm.

To illustrate this, consider the approximate change of our loss function after some parameter update with its first-order Taylor expansion:

$$\mathcal{L}(\theta + \Delta\theta) \approx \mathcal{L}(\theta) + \langle \nabla_{\theta} \mathcal{L}(\theta), \Delta\theta \rangle \quad (1)$$

When deciding on an optimizer to choose, a good optimizer would be one that minimizes our loss as much as possible with a restricted step size in our parameters (to maintain the Taylor approximation):

$$\Delta\theta^* = \arg \min_{\|\Delta\theta\| \leq \eta} \mathcal{L}(\theta + \Delta\theta) \quad (2)$$

$$\Delta\theta^* = \arg \min_{\|\Delta\theta\| \leq \eta} \mathcal{L}(\theta) + \langle \nabla_{\theta} \mathcal{L}(\theta), \Delta\theta \rangle \quad (3)$$

$$\Delta\theta^* = \arg \min_{\|\Delta\theta\| \leq \eta} \langle \nabla_{\theta} \mathcal{L}(\theta), \Delta\theta \rangle \quad (4)$$

Where η is an adjustable hyperparameter. An important observation made by [2] is that common optimizers such as SGD and Adam can be derived by changing the norm that this constraint cares about. The standard ℓ_2 -norm reduces to a scaled version of gradient descent, which can be easily adapted to SGD if we approximate our gradient using batches. Using the ℓ_{∞} -norm instead reduces to sign-GD, which closely resembles Adam if momentum is added when calculating the sign of our gradient. Muon was derived with the understanding of this theory, opting to use the RMS→RMS induced norm for matrices.

We argue that this fundamental difference in the construction Adam and Muon is a core factor that increases catastrophic forgetting when switching optimizers between pre-training and fine-tuning. When parameter updates are constrained by a particular norm, the resulting parameters live on distinct surface that minimizes the norm of the loss gradient for the given task. By later fine-tuning with a different optimizer and thus different norm constraint, the parameters no longer reside on this surface, leading to a sudden collapse of the performance in the pre-trained task.

Recent work [9] has also investigated the *sharpness* of the minima when using different optimizers, demonstrating that Adam’s implicit regularization minimizes $\text{tr}(\text{Diag}(\mathbf{H}^{1/2}))$, wherein $\mathbf{H}$ represents the Hessian matrix of the model weights.

2.3 Muon Finetuning on Adam-Pretrained Models

Recent developments have also investigated the effects of using mismatched optimizers, in this case, using Muon finetuning on Adam-pretrained models. Work by Qu et. al [14] has similarly measured the increased degradation in performance that using mismatched optimizers has on finetuning, and likewise makes assumptions about why this phenomenon may occur. They similarly propose that implicit biases made by different optimizers can affect the weights after finetuning, but they focus more on the converged solution of the model weights rather than the biases made on the change of weights from the original model. Our work extends these current investigations by specifically looking into how the weight changes are affected by different optimizer biases.

3 Method

3.1 Models and experimental design

To study optimizer mismatch and catastrophic forgetting, we fine-tune Moonlight-16B-A3B-Instruct checkpoints with matched architecture and pretraining recipes: one family trained with AdamW and one with Muon [10]. Holding these factors fixed isolates pairing of pretraining optimizer with fine-tuning optimizer when interpreting downstream benchmark retention. Through our experiments, we mainly seek to answer the following questions:

RQ1: How strongly does optimizer mismatch between pretraining and fine-tuning affect catastrophic forgetting of pretrained knowledge?

RQ2: What changes in weight geometry and loss landscape structure accompany fine-tuning under matched versus mismatched optimizers, and do these changes correlate with the degree of forgetting?

3.2 Fine-tuning Regimes

In-Distribution Fine-tuning. To isolate the effects of optimizer mismatch from pre-training to fine-tuning from the inherent shifts in the model’s loss landscape brought about by out-of-distribution fine-tuning regimes, we perform fine-tuning on AllenAI C4 [15], wherein the model is trained to continue a web-text excerpt without repeating it. Although the corpus of data utilized for pre-training of Moonlight is not publicly available, we seek to emulate typical web-scale pre-training datasets to isolate the effects of the changing loss landscape across optimizers from any confounding effects that may arise when fine-tuning with heavily out of distribution tasks. We report held-out validation metrics alongside standard downstream benchmark scores after consolidation (Section 4.1).

Out-of-Distribution Fine-tuning. To investigate catastrophic forgetting brought about by fine-tuning on specialized, out of distribution domains, we utilize a curated autonomous driving dataset as our out of distribution fine-tuning task. SimLingo [16] is a prior autonomous driving work that investigates language-action alignment when using Vision-Language-Action models for autonomous driving. The SimLingo dataset contains a set of heuristically generated driving action language labels (e.g., "Avoid the vehicle that is opening its door. Decelerate to change to the oncoming lane, as the gap is big enough"). Later work [4] augments this dataset with free-form reasoning traces describing critical agents within each autonomous driving scene. We decided to fine-tune the model to reproduce the aforementioned heuristically generated action language labels, given these reasoning traces as

input. We evaluate retention on the same benchmark suite as for C4, aggregated over checkpoints where applicable (Section 4.2).

Mitigation. We compare full-parameter updates to Low-Rank Adaptation [6] as a potential strategy for mitigation of catastrophic forgetting. We similarly measure task forgetting across matching and mismatched optimizer combinations to investigate whether optimizer mismatch can exacerbate catastrophic forgetting despite mitigation strategies such as LoRA.

3.3 Why Does Optimizer Mismatch Affect Forgetting?

To investigate the mechanisms underlying optimizer-mismatch-induced forgetting, we examine how the implicit biases of Adam and Muon manifest in the structure of their weight updates. Because each optimizer performs steepest descent under a different norm, matched and mismatched fine-tuning runs are expected to produce updates with qualitatively different geometry. To make this concrete, we measure the ℓ_∞ and RMS $\rightarrow$ RMS induced norms of the weight updates across our fine-tuning runs.

Furthermore, building upon [9], we measure sharpness $\text{tr}(\text{Diag}(\mathbf{H})^{1/2})$ via an empirical Fisher diagonal $\hat{H}_{ii} \approx \frac{1}{N} \sum_n g_i^2$ over N calibration batches from C4 [15]. As the feedforward MoE expert layers are approximately 93% of the model parameters, we focus our analysis on the MoE/MLP layers of the model.

4 Results

4.1 In-Distribution Fine-tuning (Web Text Continuation)

Table 7 reports post-fine-tuning retention (mean $\pm$ standard deviation over three random seeds) after C4 full-parameter SFT at 400 steps using **MMLU**; **GSM8K**; **TriviaQA**; **BBH**; and pass@1 **HumanEval**. Differences across optimizer pairings are small (≈ 0.1 MMLU points), consistent with in-distribution continuation staying near pretraining.

For SimLingo, Tables 8–9 pair **MMLU** with accuracy on the **C-Eval** from the LM evaluation harness bundled with Moonlight checkpoints. Multiple-choice aggregates retain discriminative separation after catastrophic full-parameter updates, whereas strict English exact-match tasks (GSM8K/BBH/HumanEval in the same harness) often register identically-zero scores once full-parameter weights are consolidated; thus, we focus our analysis on these multiple-choice selection tasks. Figure 2 plots full-parameter retention against driving-task validation exact match; Figure 3 plots **MMLU** and **CEval** versus fine-tuning step for LoRA. Numeric values for benchmark scores are available in Appendix B

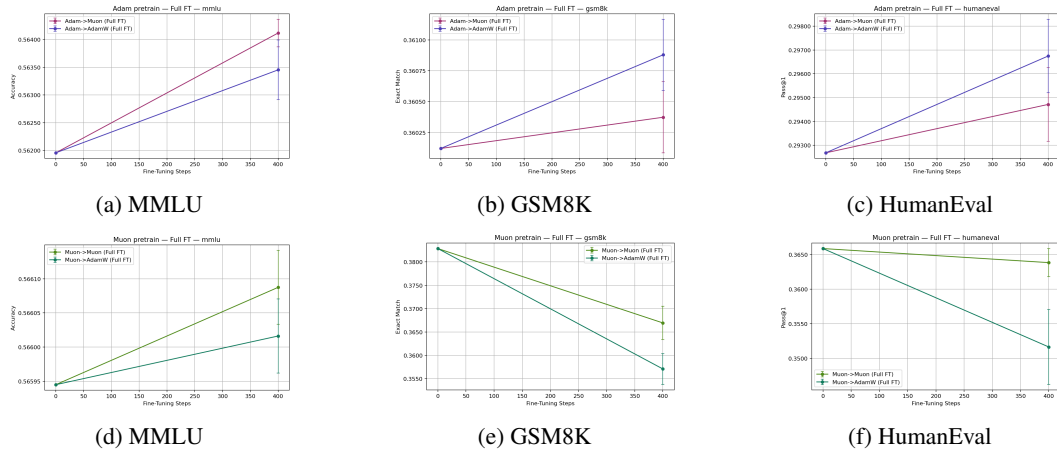


Figure 1: C4 SFT. **top** AdamW-pretrained initialization; **bottom** Muon-pretrained initialization.

4.2 Out-of-Distribution Fine-Tuning (Autonomous Driving)

Table 8 and Figure 2 summarize SimLingo full-parameter SFT. Switching AdamW-pretrained checkpoints from matched AdamW to Muon at fine-tuning moves MMLU from 29.97% to 24.84% at step 400, while paired CEval scores fall from 31.50% to 27.86%; Muon-pretrained consolidated weights stay near 23% MMLU for either fine-tuning optimizer (Table 8). Figure 2 traces each optimizer pairing as driving validation exact match increases at the cost of MMLU and C-Eval retention; the leftmost point (0% driving exact match) uses pretrained benchmark scores before any SimLingo SFT. Numeric values for benchmark scores are available in Appedix B

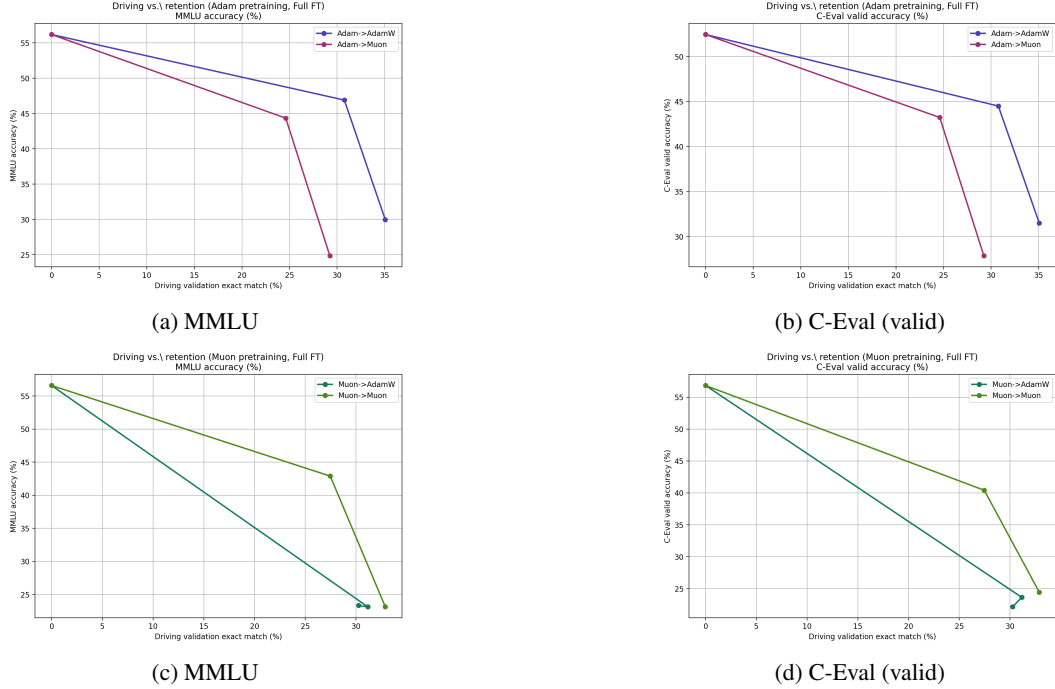


Figure 2: SimLingo full-parameter SFT: **MMLU (left)** and **C-Eval validation accuracy (right)** vs. driving-task validation exact match. Each curve is one pretrain×fine-tune optimizer pairing; step 0 uses pretrained benchmark scores with 0% driving exact match. *Top row*: AdamW-pretrained initialization; *bottom row*: Muon-pretrained.

LoRA mitigation. LoRA substantially improves retention versus full fine-tuning (Table 9, Figure 3). We report the same **MMLU/C-Eval** aggregates as Table 8. The strongest mismatch remains Muon-pretrained checkpoints fine-tuned with AdamW (30.76% MMLU and 26.82% CEval versus 55.78% / 52.82% under Muon→Muon at 2000 steps). However, fine-tuning an AdamW-pre-trained checkpoint with the Muon optimizer results in less catastrophic forgetting compared to fine-tuning with AdamW. Still, we do observe that the model pre-trained with AdamW experiences a smaller amount of catastrophic forgetting relative to the model pre-trained with Muon when both are fine-tuned with AdamW. A similar effect occurs when fine-tuning with Muon: the model pre-trained with Muon displays less catastrophic forgetting than the model pre-trained with AdamW. Numeric values for benchmark scores are available in Appendix B.

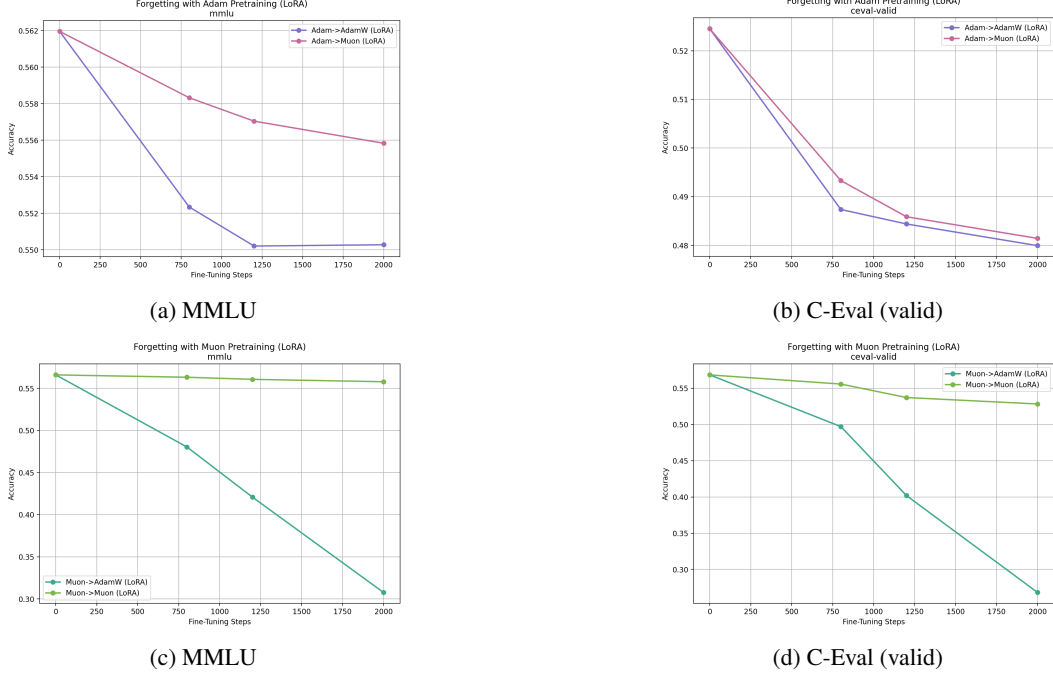


Figure 3: SimLingo LoRA SFT (2000 steps): **MMLU (left)** and **C-Eval validation accuracy (right)** vs. fine-tuning step. **Top:** AdamW-pretrained; **bottom:** Muon-pretrained.

4.3 Weight Change Analysis

For each pretrained initialization, we visualize per-tensor full fine-tuning updates by plotting ℓ_∞ versus RMS→RMS induced matrix norms for every trainable matrix, colored by structural role (attention projections versus MLP/MoE blocks).

These scatter plots give massive insight and evidence into the effects that different optimizers have on the weight changes during training. For Muon finetuning, it is clear that most weights minimize their RMS→RMS induced norm as much as possible, with there being much more variance in the ℓ_∞ norm for each matrix, as Muon does not care to minimize this norm. This causes the weights from Muon finetuning to form vertical lines in the scatter plot. The opposite is the case with AdamW finetuning, where the ℓ_∞ is minimized, leading to horizontal lines being formed.

We can interpret these plots as AdamW finetuning having the tendency to update all of the weights in the attention matrices resulting in a high RMS→RMS norm, but every individual weight not being perturbed greatly, leading to a low ℓ_∞ norm. On the other hand, Muon finetuning makes a few large shifts to a few weights resulting in high ℓ_∞ norm, but the rest of the weights in the matrices are not perturbed greatly, resulting in low RMS→RMS norm.

Out-of-distribution weight changes. Under OOD fine-tuning on SimLingo, the norm-geometry separation between optimizers is most pronounced (Figures 4 and 5). For the Adam-pretrained base, Muon-FT points (orange) in the Q and KV proj panels cluster in tight vertical bands at low RMS→RMS values (≈ 0.2) with high ℓ_∞ spread, while AdamW-FT points (blue) spread rightward along a horizontal band at lower ℓ_∞ . The O proj shows the same pattern. The MLP/MoE panel is the exception: both optimizers overlap in a shared diagonal fan, suggesting expert layers are less sensitive to norm-constraint differences—possibly because their larger parameter count makes the effective update geometry more isotropic regardless of optimizer. The Muon-pretrained base (Figure 5) reproduces the same geometry, with Muon-FT again forming vertical clusters in all attention projections while AdamW-FT spreads horizontally. An interesting observation is that this separation is most pronounced in the attention projection matrices rather than the MLP/MoE layers. The exact reason for this is unknown, but would be an interesting avenue to continue investigating for future experiments.

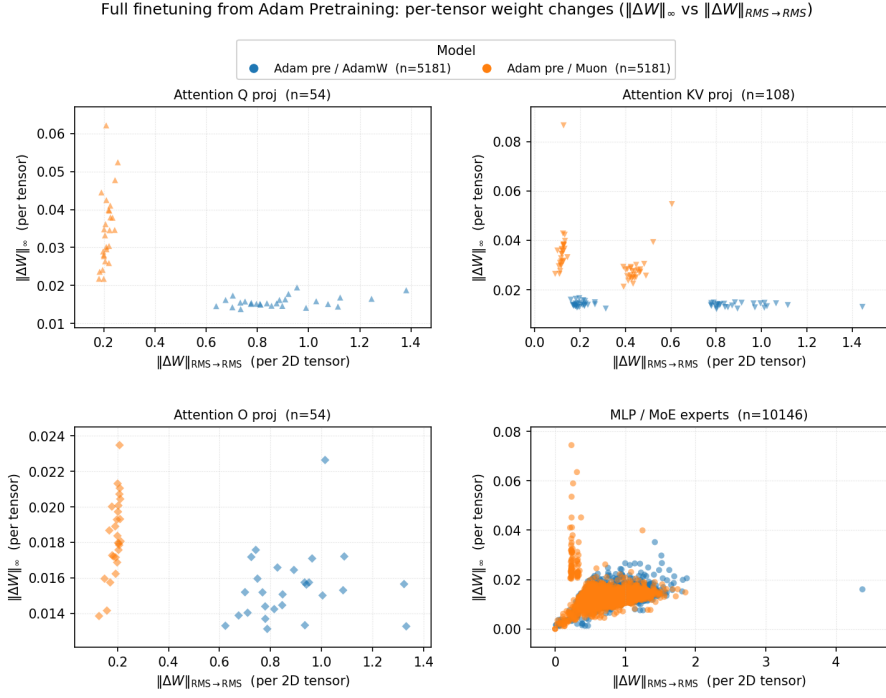


Figure 4: Per-tensor update geometry after full-parameter fine-tuning on SimLingo (ℓ_∞ vs. RMS $\rightarrow$ RMS for each matrix), colored by module type; AdamW-pretrained initialization.

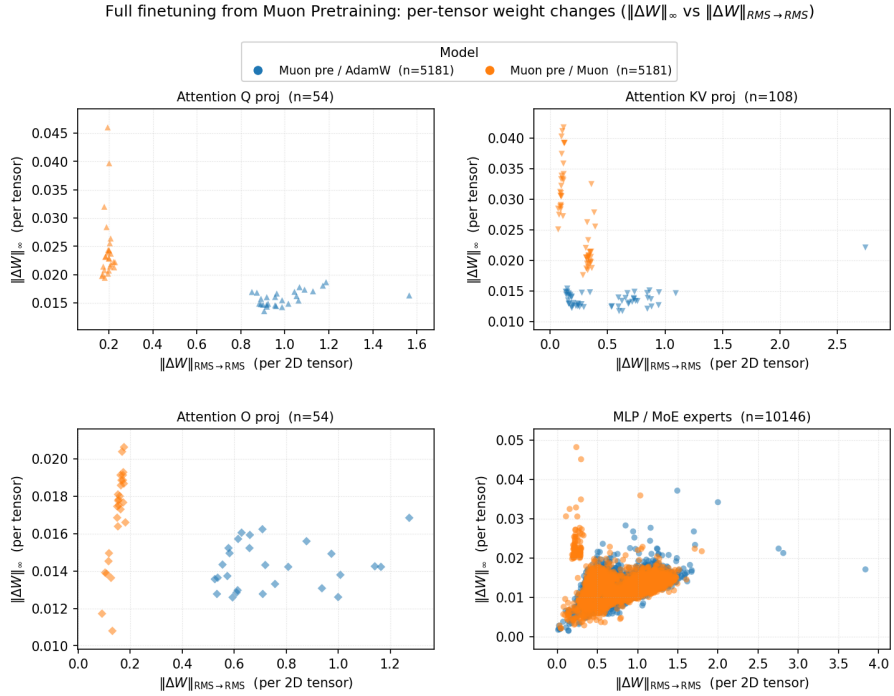


Figure 5: Same as Figure 4 with Muon-pretrained initialization.

In-distribution weight changes. The same geometric separation between AdamW and Muon weight updates is present under in-distribution C4 fine-tuning (Figures 7 and 8), though with notably smaller absolute weight-change norms overall and noticeably more overlap between optimizer conditions. Optimizers still follow qualitatively distinct loss geometries—AdamW concentrating updates along the ℓ_∞ -minimizing surface and Muon along the RMS→RMS-minimizing surface—and the attention projection matrices remain the primary site of separation while the MLP/MoE layers again show less differentiation. The reduced magnitude of weight changes relative to the OOD setting is consistent with the fine-tuning distribution lying closer to the pretraining regime, requiring smaller parameter adjustments to fit the new task.

Tensor type	AdamW LoRA	Muon LoRA	AdamW Full FT	Muon Full FT
Attention Q proj	1.149×10^{-5}	1.318×10^{-4}	1.885×10^{-6}	9.295×10^{-5}
Attention KV proj	1.225×10^{-4}	2.446×10^{-4}	9.324×10^{-7}	9.507×10^{-5}
Attention O proj	1.787×10^{-4}	9.794×10^{-6}	3.792×10^{-6}	4.785×10^{-6}
MLP / MoE experts	1.578×10^{-5}	4.705×10^{-5}	4.022×10^{-6}	9.414×10^{-6}

Table 1: AdamW pretraining: variance of $\|\Delta W\|_\infty$ across tensors in each category, by finetune variant. Bold cells mark the lower AdamW-vs-Muon variance within each finetune type (LoRA on the left of the divider, Full FT on the right).

Tensor type	AdamW LoRA	Muon LoRA	AdamW Full FT	Muon Full FT
Attention Q proj	0.1497	0.04088	0.03232	3.225×10^{-4}
Attention KV proj	0.6714	0.1189	0.1369	0.02768
Attention O proj	25.9	0.1689	0.03163	4.413×10^{-4}
MLP / MoE experts	0.3312	0.03222	0.06276	0.07066

Table 2: AdamW pretraining: variance of $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}}$ across tensors in each category, by finetune variant. Bold cells mark the lower AdamW-vs-Muon variance within each finetune type (LoRA on the left of the divider, Full FT on the right).

Tensor type	AdamW LoRA	Muon LoRA	AdamW Full FT	Muon Full FT
Attention Q proj	7.921×10^{-6}	4.947×10^{-5}	1.751×10^{-6}	3.629×10^{-5}
Attention KV proj	5.942×10^{-5}	2.959×10^{-4}	2.337×10^{-6}	5.086×10^{-5}
Attention O proj	6.136×10^{-6}	3.486×10^{-6}	1.561×10^{-6}	6.295×10^{-6}
MLP / MoE experts	1.142×10^{-5}	4.810×10^{-5}	5.190×10^{-6}	6.565×10^{-6}

Table 3: Muon pretraining: variance of $\|\Delta W\|_\infty$ across tensors in each category, by finetune variant. Bold cells mark the lower AdamW-vs-Muon variance within each finetune type (LoRA on the left of the divider, Full FT on the right).

Tensor type	AdamW LoRA	Muon LoRA	AdamW Full FT	Muon Full FT
Attention Q proj	0.2841	0.06155	0.02111	2.313×10^{-4}
Attention KV proj	0.1796	0.149	0.1808	0.01474
Attention O proj	0.08503	0.1283	0.04684	6.139×10^{-4}
MLP / MoE experts	0.5401	0.03868	0.09063	0.1008

Table 4: Muon pretraining: variance of $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}}$ across tensors in each category, by finetune variant. Bold cells mark the lower AdamW-vs-Muon variance within each finetune type (LoRA on the left of the divider, Full FT on the right).

Tables 1, 2, 3, and 4 showcase the variance of the norms of the weight updates along multiple finetuning methods and the various tensor types of the original model for the SimLingo OOD setting. These tables are here to better objectify the phenomena we see in the scatter plots of the same data, that being the clear bias between specific norms for each optimizer. The tables highlight that for

Muon finetuning, the variance in the RMS→RMS is much smaller compared to AdamW, and the opposite is true with ℓ_∞ . This is consistent over both LoRA and full finetuning.

Tables 10–13 provide the same analysis for the C4 in-distribution setting, reporting mean $\pm$ std across three seeds. The core pattern from SimLingo holds: Muon finetuning produces dramatically lower RMS→RMS variance than AdamW across all attention types while AdamW produces lower ℓ_∞ variance for Q and KV projections. Interestingly, the O projection is anomalous for ℓ_∞ : AdamW FT variance exceeds Muon FT under Adam pretraining, and similarly under Muon pretraining, a reversal of the pattern seen in Q and KV. The MLP/MoE experts again show the least separation between optimizers. Critically, the standard deviations are small relative to the means throughout, confirming that these differences are stable across seeds rather than artifacts of random variation. Full tables and weight norm visualizations are available in Appendix C

4.4 Sharpness Analysis

Following [9], we estimate $\text{tr}(\text{Diag}(\hat{H})^{1/2})$ with an empirical Fisher diagonal on C4 calibration batches, aggregated over MoE feedforward weights. We evaluate sharpness shifts only for full-parameter fine-tuning on the SimLingo dataset.

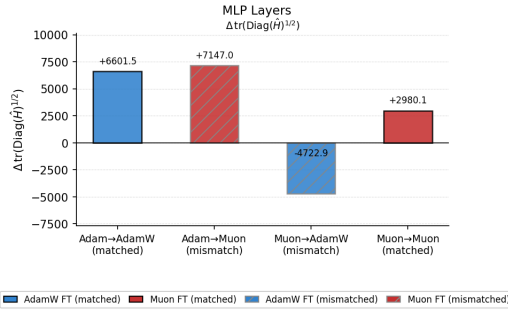


Figure 6: $\Delta \text{tr}(\text{Diag}(\hat{H})^{1/2})$ for MoE feedforward layers: **full-parameter** SimLingo SFT only (400 steps), relative to each pretrained baseline. Empirical Fisher diagonal $\hat{H}$ estimated on C4 calibration batches; four pretraining×fine-tuning optimizer pairings.

Figure 6 shows that fine-tuning with AdamW leads to a smaller increase in $\text{tr}(\text{Diag}(\hat{H})^{1/2})$ than fine-tuning with Muon for the AdamW-pretrained initialization, and that fine-tuning with AdamW *decreases* $\text{tr}(\text{Diag}(\hat{H})^{1/2})$ relative to Muon fine-tuning when starting from Muon-pretrained weights. This is consistent with AdamW’s implicit bias toward minimizing $\text{tr}(\text{Diag}(\hat{H})^{1/2})$. In the mismatched regime (AdamW fine-tuning on Muon-pretrained weights), AdamW sharply reduces $\text{tr}(\text{Diag}(\hat{H})^{1/2})$ and pulls geometry away from the structure Muon established in pretraining. We hypothesize that this mismatch in loss landscape contributes to the increased catastrophic forgetting observed under optimizer mismatch: the fine-tuning optimizer does not merely translate weights but reshapes the loss landscape in a manner less compatible with the Muon pre-trained initialization.

5 Discussion

Our results separate two axes: *optimizer mismatch* and *distribution shift*. On in-distribution C4 fine-tuning, Table 7 shows only small gaps between optimizer pairings on held-out benchmarks, so mismatch alone may be insufficient to produce large shifts when the task stays in-distribution. On out-of-distribution autonomous driving SFT, full-parameter fine-tuning yields large MMLU drops (Table 8), and LoRA recovers much of that capacity.

Limitations. Currently, our sharpness analysis only examines the full fine-tuning experiments with a far out of distribution task, and only examines the implicit sharpness minimization of AdamW. In the future, we hope to expand our sharpness and broader mechanistic analysis work to include further exploration of the implicit biases of the Muon optimizer, and further exploration of the persistence of such phenomena when fine-tuning on tasks in varying degrees of similarity to the pre-training task distribution.

Furthermore, our exploration of mitigation strategies is currently limited to LoRA. Utilizing optimizer transition schedules, additional adapter schemes, and the development of custom optimizers is a promising direction for future work towards mitigation of both catastrophic forgetting brought about by optimizer mismatch and catastrophic forgetting as a general phenomenon.

Additionally, as this is an investigation of optimizer dynamics, future work to perform deeper ablations on the effects of hyperparameter choices will provide further grounding for our results. Our current optimizer choices are based solely in fine-tuning task accuracy and pre-training task retention, but mechanistic analysis of models pre-trained with varying hyperparameters is a promising direction for future work.

Impact

Catastrophic forgetting is a fundamental challenge in continual learning: fine-tuning a model on a new task can rapidly degrade general domain knowledge and pre-training task performance. Despite its practical importance, the role of optimizer choice in driving this phenomenon is still poorly understood. In this work, we shed light on this phenomenon, revealing where, how, and why it occurs. By systematically varying the optimizer used during pre-training and fine-tuning, we show that mismatched optimizers measurably worsen forgetting, and we provide evidence for why: different optimizers impose different implicit norm constraints on weight updates, pushing parameters off the surface shaped by pre-training. To enable others seeking to investigate catastrophic forgetting to build upon our work, we have open-sourced our mechanistic analysis code at <https://github.com/swetha2022/194-final-project>.

Study of catastrophic forgetting matters especially in the LLM setting, where models containing billions of parameters are frequently adapted to new domains through instruction tuning, domain adaptation, or alignment procedures. Unchecked forgetting can degrade general reasoning ability, factual knowledge, or safety alignment when optimizing for narrow objectives. Understanding the optimizer dynamics that contribute to forgetting and demonstrating their effects under controlled conditions is a necessary step toward more stable training pipelines and better retention of pre-trained knowledge. Our work makes a concrete effort in this direction: we isolate optimizer mismatch as a measurable driver of forgetting, and trace it back to fundamental differences in how optimizers constrain weight updates, laying the groundwork for principled, optimizer-aware fine-tuning practices.

Acknowledgments

We thank course staff and our classmates for their feedback. We also thank NVIDIA for sponsoring our compute, enabling us to be able to carry out these experiments.

SimLingo training logs are available at <https://wandb.ai/cs194-squad/moonlight-driving-ft>. C4 training logs are available at <https://wandb.ai/katiawang-university-of-california-berkeley/sft-dev>. Our code is available at <https://github.com/swetha2022/194-final-project> and our blogpost can be viewed at <https://swetha2022.github.io/cs194/>.

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A Hyperparameters

Hyperparameter	LoRA+AdamW	LoRA+Muon	FT+AdamW	FT+Muon
<i>SimLingo SFT</i>				
Steps	2000	2000	400	400
Batch size	64	64	64	64
Learning rate	2×10^{-4}	2×10^{-4}	2×10^{-4}	2×10^{-4}
Weight decay	0.01	0.01	0.01	0.01
Muon momentum	–	0.95	–	0.95
LR schedule	Cosine	Cosine	Cosine	Cosine

Table 5: SimLingo fine-tuning hyperparameters (shared across optimizer cells unless noted).

Hyperparameter	LoRA	Full FT
<i>C4 SFT</i>		
Steps	2000	400
Global batch size	8	8
Max sequence length	2048	2048
LoRA rank / α	16 / 32	—
LoRA dropout	0.05	—
Finetune LR (AdamW)	10^{-5}	10^{-5}
Finetune LR (Muon)	10^{-5}	10^{-5}
Weight decay	0.01	0.01

Table 6: C4 fine-tuning hyperparameters.

B Numeric Benchmark Scores for Fine-tuning Experiments

Pre	FT	MMLU	GSM8K	TriviaQA	BBH	HumanEval
AdamW	—	56.20	36.01	48.98	46.69	29.27
AdamW	AdamW	56.35 ± 0.09	36.09 ± 0.33	49.02 ± 0.12	46.95 ± 0.38	29.67 ± 0.35
AdamW	Muon	56.41 ± 0.04	36.04 ± 0.24	48.64 ± 0.09	46.97 ± 0.31	29.47 ± 0.35
Muon	—	56.59	38.29	53.48	46.46	36.59
Muon	AdamW	56.60 ± 0.08	35.71 ± 0.57	53.22 ± 0.25	46.01 ± 0.09	35.16 ± 1.27
Muon	Muon	56.61 ± 0.07	36.69 ± 0.62	53.15 ± 0.14	46.57 ± 0.25	36.38 ± 0.35

Table 7: C4 fine-tuned model task accuracy (mean $\pm$ std.) over three seeds

Pre	FT	MMLU	CEval (valid)
AdamW	—	56.20	52.45
AdamW	AdamW	29.97	31.50
AdamW	Muon	24.84	27.86
Muon	—	56.59	56.84
Muon	AdamW	23.32	22.14
Muon	Muon	23.13	24.44

Table 8: Benchmark Accuracies for SimLingo full-parameter SFT.

Pre	FT	MMLU	CEval (valid)
AdamW	—	56.20	52.45
AdamW	AdamW	55.03	47.99
AdamW	Muon	55.58	48.14
Muon	—	56.59	56.84
Muon	AdamW	30.76	26.82
Muon	Muon	55.78	52.82

Table 9: SimLingo LoRA SFT (2000 steps), matching the metric layout of Table 8. Rows list merged adapters evaluated once per pairing.

C Weight Analysis for C4 Full Fine-tuning Experiments

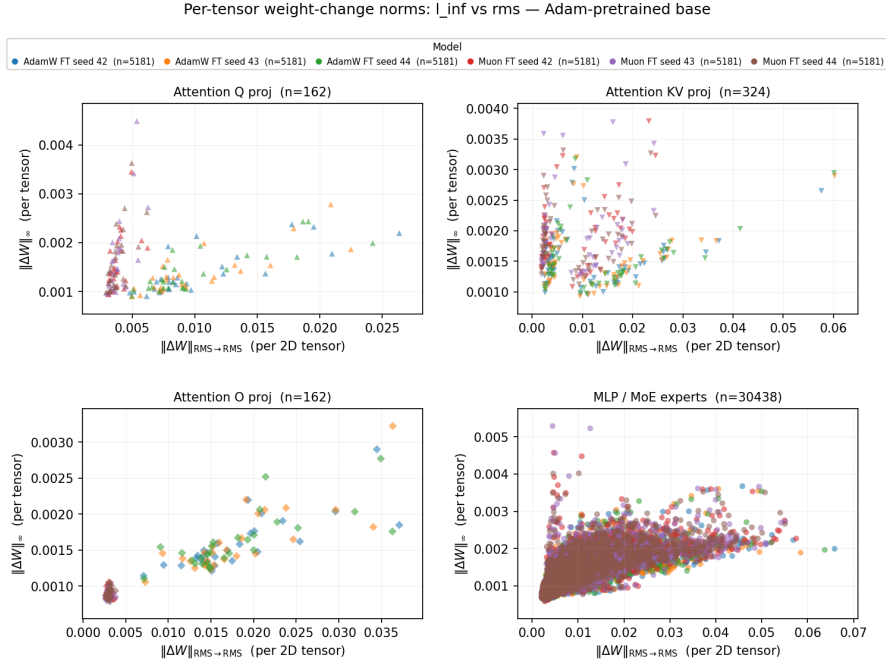


Figure 7: Per-tensor update geometry after full-parameter fine-tuning on C4 (ℓ_∞ vs. RMS→RMS for each matrix), colored by module type; AdamW-pretrained initialization.

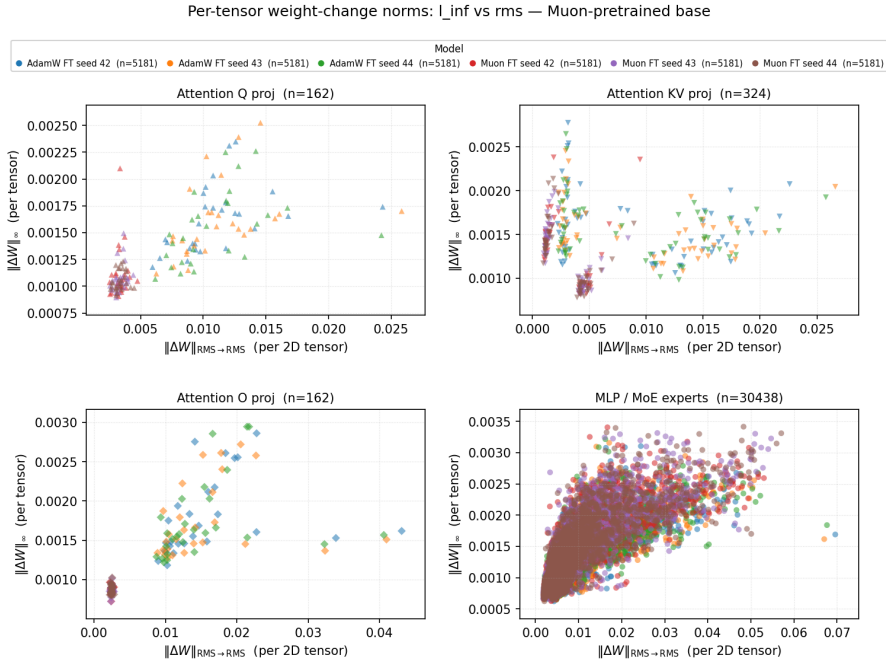


Figure 8: Same as Figure 7 with Muon-pretrained initialization.

Tensor type	AdamW Full FT	Muon Full FT
Attention Q proj	$1.902 \times 10^{-7} \pm 8.463 \times 10^{-9}$	$4.698 \times 10^{-7} \pm 1.735 \times 10^{-7}$
Attention KV proj	$2.195 \times 10^{-7} \pm 1.559 \times 10^{-8}$	$3.059 \times 10^{-7} \pm 5.461 \times 10^{-8}$
Attention O proj	$1.598 \times 10^{-7} \pm 3.024 \times 10^{-8}$	$3.446 \times 10^{-9} \pm 8.612 \times 10^{-10}$
MLP / MoE experts	$1.124 \times 10^{-7} \pm 4.409 \times 10^{-9}$	$1.838 \times 10^{-7} \pm 3.192 \times 10^{-9}$

Table 10: AdamW pretraining: mean $\pm$ std across runs of the category variance of $\|\Delta W\|_{\infty}$. Bold cells indicate cases where AdamW achieves lower variance than Muon under a full fine-tuning setup.

Tensor type	AdamW Full FT	Muon Full FT
Attention Q proj	$2.426 \times 10^{-5} \pm 3.352 \times 10^{-6}$	$6.141 \times 10^{-7} \pm 4.364 \times 10^{-8}$
Attention KV proj	$1.242 \times 10^{-4} \pm 4.528 \times 10^{-6}$	$4.579 \times 10^{-5} \pm 1.269 \times 10^{-7}$
Attention O proj	$4.837 \times 10^{-5} \pm 2.013 \times 10^{-6}$	$4.256 \times 10^{-8} \pm 3.122 \times 10^{-9}$
MLP / MoE experts	$3.298 \times 10^{-5} \pm 7.077 \times 10^{-7}$	$4.521 \times 10^{-5} \pm 7.626 \times 10^{-7}$

Table 11: AdamW pretraining: mean $\pm$ std across runs of the category variance of $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}}$. Bold cells indicate cases where AdamW achieves lower variance than Muon under a full fine-tuning setup.

Tensor type	AdamW Full FT	Muon Full FT
Attention Q proj	$1.103 \times 10^{-7} \pm 1.469 \times 10^{-8}$	$2.815 \times 10^{-8} \pm 2.626 \times 10^{-8}$
Attention KV proj	$1.187 \times 10^{-7} \pm 2.761 \times 10^{-8}$	$1.286 \times 10^{-7} \pm 1.320 \times 10^{-8}$
Attention O proj	$2.394 \times 10^{-7} \pm 1.919 \times 10^{-8}$	$2.995 \times 10^{-9} \pm 3.281 \times 10^{-10}$
MLP / MoE experts	$1.284 \times 10^{-7} \pm 1.178 \times 10^{-9}$	$2.124 \times 10^{-7} \pm 3.269 \times 10^{-9}$

Table 12: Muon pretraining: mean $\pm$ std across runs of the category variance of $\|\Delta W\|_{\infty}$. Bold cells indicate cases where AdamW achieves lower variance than Muon under a full fine-tuning setup.

Tensor type	AdamW Full FT	Muon Full FT
Attention Q proj	$1.439 \times 10^{-5} \pm 7.753 \times 10^{-7}$	$2.146 \times 10^{-7} \pm 2.064 \times 10^{-8}$
Attention KV proj	$4.189 \times 10^{-5} \pm 8.439 \times 10^{-7}$	$3.530 \times 10^{-6} \pm 2.169 \times 10^{-7}$
Attention O proj	$5.525 \times 10^{-5} \pm 3.717 \times 10^{-6}$	$1.501 \times 10^{-8} \pm 4.242 \times 10^{-10}$
MLP / MoE experts	$3.058 \times 10^{-5} \pm 4.136 \times 10^{-7}$	$4.213 \times 10^{-5} \pm 7.744 \times 10^{-7}$

Table 13: Muon pretraining: mean $\pm$ std across runs of the category variance of $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}}$. Bold cells indicate cases where AdamW achieves lower variance than Muon under a full fine-tuning setup.